

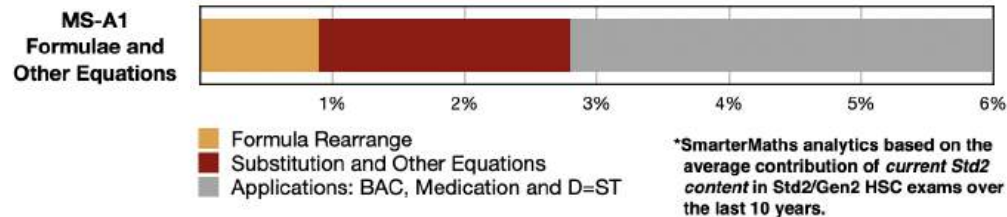
STD 1: Algebra (Std 1), A1 Formulae and Equations (Y11)

Formula Rearrange (Std 1)

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Exam Equivalent Time: 43.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

Gen2/Std2 Exam Contribution History A1 Formulae and Equations



IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- MS-A1 *Formulae and Equations* has accounted for 6% and 5% of the 2020 and 2019 Std1 exams respectively.
- MS-A1 *Formulae and Equation* is split into the sub-topics listed in the graph above. Note that estimated exam contributions in the graph are taken from Std2/Gen2 past exams due to the very limited Std1 exam history. They should only be used for estimating the "relative" importance of each sub-topic.
- This analysis looks at the sub-topic *Formula Rearrange*.

ANALYSIS - What to Expect and Common pitfalls

- Although not tested in the 2020 Std1 HSC, a substantial 3-mark question appeared in the 2019 Std1 exam, producing a surprisingly low mean mark of 32%.
- We note that students sitting the 2021 Std1 HSC can be required to rearrange both linear and non-linear equations (the latter is expanded content applicable from 2021 onwards).
- Formula rearranging presents very achievable marks for most students but be careful, as some examples (in both linear and non-linear equations) have caused problems in the past and deserve attention (see 2019 Std2 11 MC, 2017 Std2 28d and 2013 Std2 21 MC).

Questions

1. Algebra, STD2 A1 2010 HSC 18 MC

Which of the following correctly express x as the subject of $a = \frac{nx}{5}$?

- (A) $x = \frac{an}{5}$
- (B) $x = \frac{5a}{n}$
- (C) $x = \frac{a-5}{n}$
- (D) $x = 5a - n$

2. Algebra, STD2 A1 2012 HSC 21 MC

Which of the following correctly expresses c as the subject of $E = mc^2 + p$?

- (A) $c = \pm \sqrt{\frac{E-p}{m}}$
- (B) $c = \pm \frac{\sqrt{E-p}}{m}$
- (C) $c = \pm \sqrt{\frac{E}{m} - p}$
- (D) $c = \pm \sqrt{\frac{E}{m} - p}$

3. Algebra, STD2 A1 2004 HSC 11 MC

If $d = 6t^2$, what is a possible value of t when $d = 2400$?

- (A) 0.05
- (B) 20
- (C) 120
- (D) 400

4. Algebra, STD2 A1 2016 HSC 24 MC

Which of the following correctly expresses Q as the subject of $e = iR + \frac{Q}{C}$?

(A) $Q = Ce + CiR$

(B) $Q = Ce - CiR$

(C) $Q = \frac{e + iR}{C}$

(D) $Q = \frac{e - iR}{C}$

5. Algebra, STD2 A1 2011 HSC 18 MC

Which of the following correctly expresses a as the subject of $s = ut + \frac{1}{2}at^2$?

(A) $a = \frac{2(s - ut)}{t^2}$

(B) $a = \frac{2s - ut}{t^2}$

(C) $a = \frac{\frac{1}{2}(s - ut)}{t^2}$

(D) $a = \frac{\frac{1}{2}s - ut}{t^2}$

6. Algebra, STD2 A1 2013 HSC 21 MC

Which equation correctly shows r as the subject of $S = 800(1 - r)$?

(A) $r = \frac{800 - S}{800}$

(B) $r = \frac{S - 800}{800}$

(C) $r = 800 - S$

(D) $r = S - 800$

7. Algebra, STD2 A1 2006 HSC 18 MC

What is the formula for q as the subject of $4p = 5t + 2q^2$?

(A) $q = \pm \frac{\sqrt{5t - 4p}}{2}$

(B) $q = \pm \frac{\sqrt{4p + 5t}}{2}$

(C) $q = \pm \sqrt{\frac{5t - 4p}{2}}$

(D) $q = \pm \sqrt{\frac{4p - 5t}{2}}$

8. Algebra, STD2 A1 2007 HSC 19 MC

Which of the following correctly expresses T as the subject of $B = 2\pi\left(R + \frac{T}{2}\right)$?

(A) $T = \frac{B}{\pi} - 2R$

(B) $T = \frac{B}{\pi} - R$

(C) $T = 2R - \frac{B}{\pi}$

(D) $T = \frac{B}{4\pi} - \frac{R}{2}$

9. Algebra, STD2 A1 2019 HSC 11 MC

Which of the following correctly expresses y as the subject of the formula $3x - 4y - 1 = 0$

A. $y = \frac{3}{4}x - 1$

B. $y = \frac{3}{4}x + 1$

C. $y = \frac{3x - 1}{4}$

D. $y = \frac{3x + 1}{4}$

10. Algebra, STD2 A1 SM-Bank 9

The volume of a sphere is given by $V = \frac{4}{3}\pi r^3$ where r is the radius of the sphere.

If the volume of a sphere is 220 cm^3 , find the radius, to 1 decimal place. (3 marks)

11. Algebra, STD2 A1 2015 HSC 28d

The formula $C = \frac{5}{9}(F - 32)$ is used to convert temperatures between degrees Fahrenheit (F) and degrees Celsius (C).

Convert 3°C to the equivalent temperature in Fahrenheit. (2 marks)

12. Algebra, STD2 A1 SM-Bank 6

Make p the subject of the equation $c = \frac{5}{3}p + 15$. (2 marks)

13. Algebra, STD2 A1 SM-Bank 11

Make r the subject of the equation $V = \frac{4}{3}\pi r^3$. (3 marks)

14. Algebra, STD2 A1 SM-Bank 12

Make F the subject of the equation $C = \frac{5}{9}(F - 32)$. (2 marks)

15. Algebra, STD2 A1 2005 HSC 24c

Make L the subject of the equation $T = 2\pi L^2$. (2 marks)

16. Algebra, STD2 A1 2017 HSC 28d

Make y the subject of the equation $x = \sqrt{yp - 1}$. (2 marks)

17. Algebra, STD1 A1 2019 HSC 34

Given the formula $C = \frac{A(y + 1)}{24}$, calculate the value of y when $C = 120$ and $A = 500$. (3 marks)

18. Algebra, STD1 A1 2021 HSC 16

Make r the subject of the formula $P = 2r + 10$. (2 marks)

Worked Solutions

1. Algebra, STD2 A1 2010 HSC 18 MC

$$a = \frac{nx}{5}$$

$$nx = 5a$$

$$x = \frac{5a}{n}$$

$\Rightarrow B$

2. Algebra, STD2 A1 2012 HSC 21 MC

$$E = mc^2 + p$$

$$mc^2 = E - p$$

$$c^2 = \frac{E - p}{m}$$

$$\therefore c = \pm \sqrt{\frac{E - p}{m}}$$

$\Rightarrow A$

3. Algebra, STD2 A1 2004 HSC 11 MC

$$d = 6t^2$$

$$t^2 = \frac{d}{6}$$

$$t = \pm \sqrt{\frac{d}{6}}$$

When $d = 2400$

$$t = \pm \sqrt{\frac{2400}{6}}$$

$$= \pm 20$$

$\Rightarrow B$

Worked Solutions

4. Algebra, STD2 A1 2016 HSC 24 MC

$$e = iR + \frac{Q}{C}$$

$$\frac{Q}{C} = e - iR$$

$$\begin{aligned}\therefore Q &= C(e - iR) \\ &= Ce - CiR\end{aligned}$$

$\Rightarrow B$

5. Algebra, STD2 A1 2011 HSC 18 MC

$$s = ut + \frac{1}{2}at^2$$

$$\frac{1}{2}at^2 = s - ut$$

$$at^2 = 2(s - ut)$$

$$a = \frac{2(s - ut)}{t^2}$$

$\Rightarrow A$

6. Algebra, STD2 A1 2013 HSC 21 MC

$$S = 800(1 - r)$$

$$1 - r = \frac{S}{800}$$

$$r = 1 - \frac{S}{800}$$

$$= \frac{800 - S}{800}$$

$\Rightarrow A$

◆◆ Mean mark 27%

7. Algebra, STD2 A1 2006 HSC 18 MC

$$4p = 5t + 2q^2$$

$$2q^2 = 4p - 5t$$

$$q^2 = \frac{4p - 5t}{2}$$

$$q = \pm \sqrt{\frac{4p - 5t}{2}}$$

$$\Rightarrow D$$

8. Algebra, STD2 A1 2007 HSC 19 MC

$$B = 2\pi \left(R + \frac{T}{2} \right)$$

$$\frac{B}{2\pi} = R + \frac{T}{2}$$

$$\frac{T}{2} = \frac{B}{2\pi} - R$$

$$T = \frac{B}{\pi} - 2R$$

$$\Rightarrow A$$

9. Algebra, STD2 A1 2019 HSC 11 MC

$$3x - 4y - 1 = 0$$

$$4y = 3x - 1$$

$$\therefore y = \frac{3x - 1}{4}$$

$$\Rightarrow C$$

♦ Mean mark 50%.

10. Algebra, STD2 A1 SM-Bank 9

$$V = \frac{4}{3}\pi r^3$$

$$3V = 4\pi r^3$$

$$r^3 = \frac{3V}{4\pi}$$

$$\text{When } V = 220$$

$$r^3 = \frac{3 \times 220}{4\pi}$$

$$= 52.521\dots$$

$$\therefore r = \sqrt[3]{52.521\dots}$$

$$= 3.744\dots \text{ (by calc)}$$

$$= 3.7 \text{ cm (to 1 d.p.)}$$

11. Algebra, STD2 A1 2015 HSC 28d

$$C = \frac{5}{9}(F - 32)$$

$$F - 32 = \frac{9}{5}C$$

$$F = \frac{9}{5}C + 32$$

$$\text{When } C = 3,$$

$$F = \left(\frac{9}{5} \times 3 \right) + 32$$

$$= 37.4 \text{ degrees } F$$

12. Algebra, STD2 A1 SM-Bank 6

$$\begin{aligned}c &= \frac{5}{3}p + 15 \\ \frac{5}{3}p &= c - 15 \\ p &= \frac{3}{5}(c - 15) \\ &= \frac{3}{5}c - 9\end{aligned}$$

13. Algebra, STD2 A1 SM-Bank 11

$$\begin{aligned}V &= \frac{4}{3}\pi r^3 \\ 3V &= 4\pi r^3 \\ \frac{3V}{4} &= \pi r^3 \\ r^3 &= \frac{3V}{4\pi} \\ r &= \sqrt[3]{\frac{3V}{4\pi}}\end{aligned}$$

14. Algebra, STD2 A1 SM-Bank 12

$$\begin{aligned}C &= \frac{5}{9}(F - 32) \\ 9C &= 5(F - 32) \\ \frac{9C}{5} &= F - 32 \\ \therefore F &= \frac{9C}{5} + 32\end{aligned}$$

15. Algebra, STD2 A1 2005 HSC 24c

$$\begin{aligned}T &= 2\pi L^2 \\ L^2 &= \frac{T}{2\pi} \\ \therefore L &= \pm \sqrt{\frac{T}{2\pi}}\end{aligned}$$

16. Algebra, STD2 A1 2017 HSC 28d

$$\begin{aligned}x &= \sqrt{yp - 1} \\ yp - 1 &= x^2 \\ yp &= x^2 + 1 \\ \therefore y &= \frac{x^2 + 1}{p}\end{aligned}$$

♦ Mean mark 46%.

17. Algebra, STD1 A1 2019 HSC 34

Make y the subject:

$$\begin{aligned}C &= \frac{A(y + 1)}{24} \\ 24C &= A(y + 1) \\ y + 1 &= \frac{24C}{A} \\ y &= \frac{24C}{A} - 1 \\ &= \frac{24 \times 120}{500} - 1 \\ &= 4.76\end{aligned}$$

♦♦ Mean mark 32% (Std1).

18. Algebra, STD1 A1 2021 HSC 16

$$\begin{aligned}P &= 2r + 10 \\ 2r &= P - 10 \\ r &= \frac{P - 10}{2}\end{aligned}$$

♦♦ Mean mark 29%.